

4.08 Further Calculus				
4.08a	Maclaurin series		a) Be able to find the Maclaurin series of a function, including the general term.	D5 D6
4.08b			b) Recognise and be able to use the Maclaurin series for e^x , $\sin x$, $\cos x$, $\ln(1+x)$ and $(1+x)^n$, and functions based on these. <i>The interval of validity should be understood.</i> <i>[Proof of the interval of validity and the use of non-real values of x are excluded.]</i>	
4.08c	Improper integrals		c) Be able to evaluate improper integrals where either the integrand is undefined at a value in the range of integration or where the range of integration is infinite. <i>e.g. $\int_0^2 \frac{1}{\sqrt{x}} dx$ or $\int_1^\infty \frac{1}{x^2} dx$.</i>	E1

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...	DfE Ref.
4.08d	Volumes of solids of revolution		d) Be able to derive formulae for and calculate volumes of solids of revolution. <i>To include solids generated using either coordinate axis as the axis of rotation, and the volume of a solid formed by rotation of a region between two curves.</i> <i>This includes curves defined parametrically.</i>	E2
4.08e	Mean values		e) Understand and be able to evaluate the mean value of a function. <i>Includes the use of: mean value = $\frac{1}{b-a} \int_a^b f(x) dx$.</i>	E3
4.08f	Partial fractions		f) Be able to integrate using partial fractions. <i>See Further Algebra section 4.05c and H240 section 1.02y for permitted forms.</i>	E4
4.08g	Inverse trigonometric and hyperbolic functions		g) Be able to derive and use the derivatives of $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$, $\sinh^{-1} x$, $\cosh^{-1} x$ and $\tanh^{-1} x$.	E5
4.08h	Further integration		h) Be able to integrate functions of the form: $\frac{1}{\sqrt{a^2 - x^2}}, \frac{1}{a^2 + x^2}, \frac{1}{\sqrt{x^2 - a^2}} \text{ and } \frac{1}{\sqrt{x^2 + a^2}}$ and use an appropriate inverse trigonometric or hyperbolic substitution for the evaluation of associated definite or indefinite integrals.	E6 H5